

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Placebo-controlled Study to Evaluate the Safety, Tolerability and Pharmacokinetics of Single and Multiple Ascending Doses of LTP001 in Healthy Adult Participants (Part A) and to Evaluate the Efficacy and Safety of LTP001 for the Treatment of Participants With Pulmonary Arterial Hypertension (Part B) **Short Title/Acronym:** A Study to Learn About the Treatment LTP001 in Healthy Participants and in Participants With PAH **Protocol Number:** CLTP001A12202 **NCT Number:** NCT06649110 **Sponsor Name:** Novartis **Investigational Product:** LTP001 (SMURF1 inhibitor) **Phase of Development:** Phase I/II **Medical Monitor:** Novartis Clinical Operations Lead / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and pharmacokinetics of single and multiple ascending doses of LTP001 in healthy adult participants (Part A) and to evaluate the efficacy (change in pulmonary vascular resistance) and safety of LTP001 in participants with Pulmonary Arterial Hypertension (Part B). **Secondary Objectives:** To assess changes in the 6-minute walk distance (6MWD), WHO functional class, and additional pharmacokinetic endpoints (AUC, Tmax, T1/2).

2.2 Background & Rationale To address the need for new disease-modifying treatments in Pulmonary Arterial Hypertension (PAH). While standard-of-care therapies exist, PAH remains a progressive condition with vascular remodeling. This study investigates LTP001, a SMURF1 inhibitor, on top of stable background therapies to assess its safety and efficacy in halting or reversing PAH progression.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 232 participants **Number of Sites:** United States + 11 other countries **Estimated Study Start:** 24-Oct-2024 **Estimated Primary Completion:** 31-Dec-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 100 years. 2. **Part A:** Healthy males and non-child-bearing potential females. 3. **Part B:** Confirmed diagnosis of PAH, pre-randomization PVR ≥ 400 dyn·sec·cm⁻⁵, treatment with stable doses of standard-of-care PAH therapies, and a 6-minute walk distance (6MWD) ≥ 150 m and ≤ 450 m.

4.2 Exclusion Criteria 1. Clinically significant ECG or cardiac abnormalities, or any surgical/medical condition which might significantly alter the absorption, distribution, metabolism, or excretion of drugs. 2. Women of child-bearing potential unless using highly effective methods of contraception. 3. Sexually active males unwilling to use a condom during intercourse while taking study treatment and for 24 hours after stopping study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single and multiple ascending doses (Part A); Therapeutic dosing regimen (Part B). **Route of Administration:** Investigational Administration (e.g., Oral/SC). **Duration of Treatment:** Up to 4 years (through completion of Part B and safety extension).

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of standard-of-care PAH therapies. **Prohibited:** Medications that could significantly alter the absorption, distribution, metabolism, or excretion of the investigational drug.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, ECG, Baseline Labs
Baseline	Day 0	Randomization, First Dose, Right Heart Catheterization (Part B), 6MWD
Interim	Week [X] to [Y]	Safety Labs, PK Sampling (\$Cmax\$, \$Tmax\$, \$AUC\$), Efficacy Assessment
End of Study	Week [X]	Final Vital Signs, Exit Interview, IP Accountability, Safety Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: **Part A:** Safety, tolerability, and peak plasma concentration (\$Cmax\$). **Part B:** Change in pulmonary vascular resistance (PVR) from baseline. **Measurement Method:** PK blood sampling analysis (Part A) and Right Heart Catheterization (Part B).

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in 6-minute walk distance (6MWD). 2. Change from baseline in WHO functional class, AUClast, AUC, $T_{1/2}$, and treatment-emergent adverse events.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of treatment-emergent adverse events, vital signs, ECG, and clinical laboratory evaluations up to the end-of-study visit. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External safety monitoring board assigned by Novartis.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes; Safety Set for adverse events and PK parameters. **Sample Size Justification:** $N=232$ adequately powered ($\alpha = 0.05$, $Power = 80\%$) to detect a significant change in PVR and 6MWD versus placebo. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) and Multiple Imputation (MI) techniques applied to continuous efficacy endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote monitoring visits for source data verification. **Audit Trail:** Standardized eCRF locking processes and electronic data clarification forms.

11. Study Identifiers & Governance

Primary ID: CLTP001A12202 **Registry Number:** NCT06649110 **Sponsor/Lead Organization:** Novartis **Study Title:** A Study to Learn About the Treatment LTP001 in Healthy Participants (Part A) and in Participants With PAH (Part B) **Official Regulatory Title:** A Randomized, Double-blind, Placebo-controlled Study to Evaluate the Safety, Tolerability and Pharmacokinetics of Single and Multiple Ascending Doses of LTP001 in Healthy Adult Participants (Part A) and to Evaluate the Efficacy and Safety of LTP001 for the Treatment of Participants With Pulmonary Arterial Hypertension (Part B)

12. Clinical Framework (PAH Specific)

Primary Condition: Pulmonary Arterial Hypertension (PAH)
Diagnosis Threshold: Pre-randomization PVR ≥ 400 dyn·sec·cm² and 6MWD between 150 m and 450 m
Medical Methodology: Stable Standard-of-care PAH Therapies
Optimization Target: Pulmonary vascular resistance (PVR) reduction and 6MWD improvement

13. Study Procedures & Methodology

Management Pathway: Standard clinical testing for PAH (Right Heart Catheterization, 6MWD)
Education Protocol (HCP): Protocol-specific training for right heart catheterization and 6MWD standardization
Education Protocol (Patient): Standardized instructions and encouragement for self-paced 6MWD testing
Quality Audit Mechanism: Centralized hemodynamic and ECG reading; clinical site monitoring

14. Observation & Follow-Up Schedule

Total Duration: Follow-up through trial completion (up to Dec 2028)
Onsite Visit Cadence: Regular intervals for PK profiling, hemodynamic assessments, and 6MWD
Remote Monitoring Frequency: Every 4 weeks (Q4W) or as per protocol design
Checklist Review Interval: End-of-study evaluation and safety extension tracking

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				